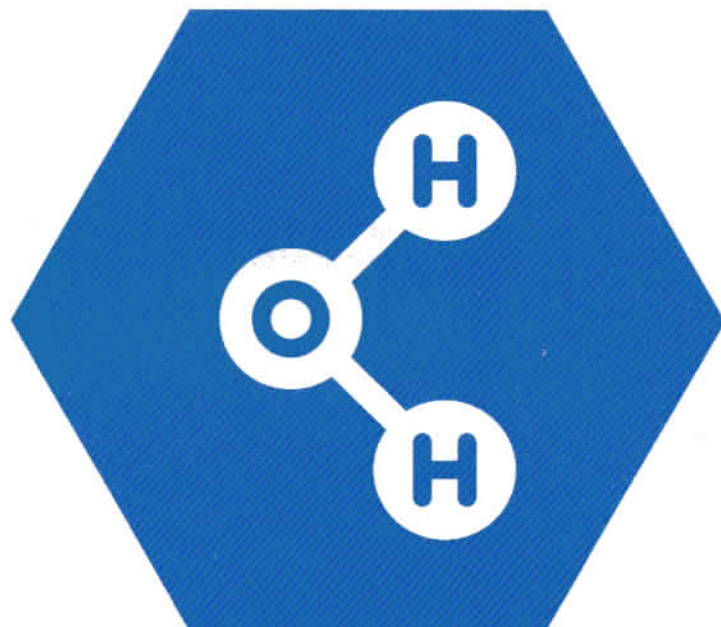




UNSW Global
AUSTRALIA

PAPER C



2016 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

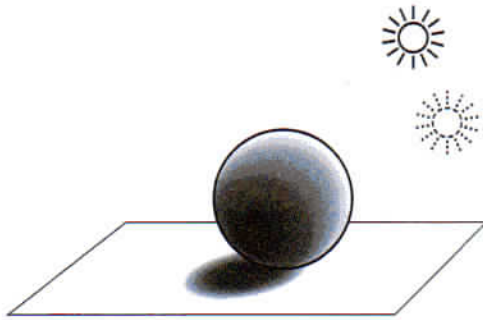
You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. The drawing shows the Sun shining on a ball on a sheet of paper.

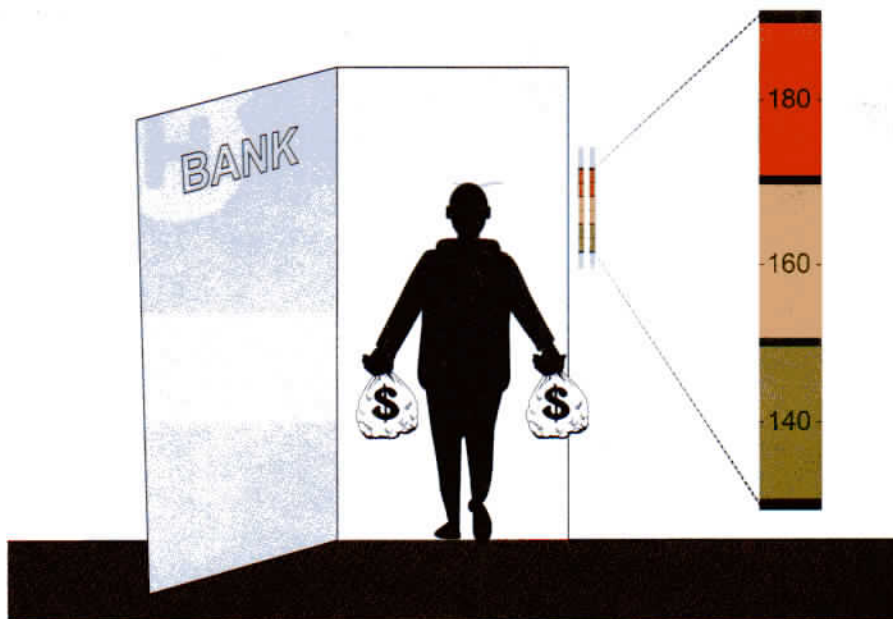


What will happen to the shadow of the ball as the Sun moves to the dotted position?

The shadow of the ball will

- (A) become longer. (B) become shorter.
(C) become rounder. (D) disappear.

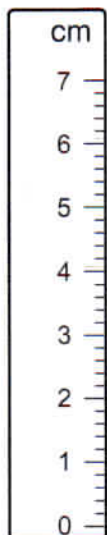
2. The diagram represents a security camera photo of a robber leaving a bank. Beside the door is a coloured strip of tape used to measure height.



Approximately how many centimetres tall was the robber?

- (A) 140
(B) 160
(C) 180
(D) 190

3. Milan had two identical candles. Milan lit candle A and let it burn for one hour. He then blew the flame out. Milan measured the height of the candles with a ruler.



candle A



candle B



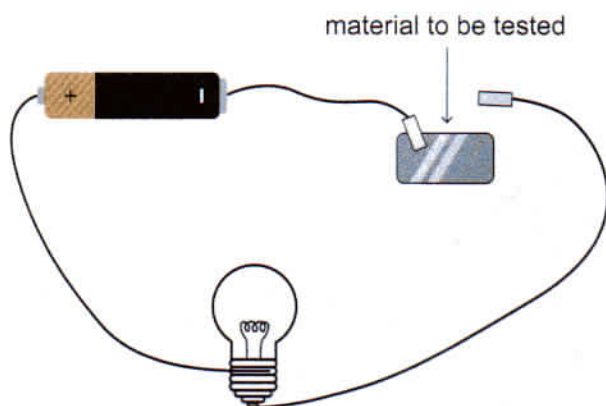
Milan then lit candle B.

How long will it take for candle B to burn completely?

- (A) 2 hours
(B) 3 hours
(C) 4 hours
(D) 6 hours

4. A material that allows electricity to flow through it is called a conductor. A material that does not allow electricity to flow through is called an insulator.

Joshua used a lamp, a battery and some wires to test some materials to see which ones are conductors. He set up his investigation as shown and summarised his results in the table.



Material tested		
glass		✓
iron	✓	
plastic		✓
wood		✓
aluminium	✓	
tin	✓	

Which list contains only materials that are insulators?

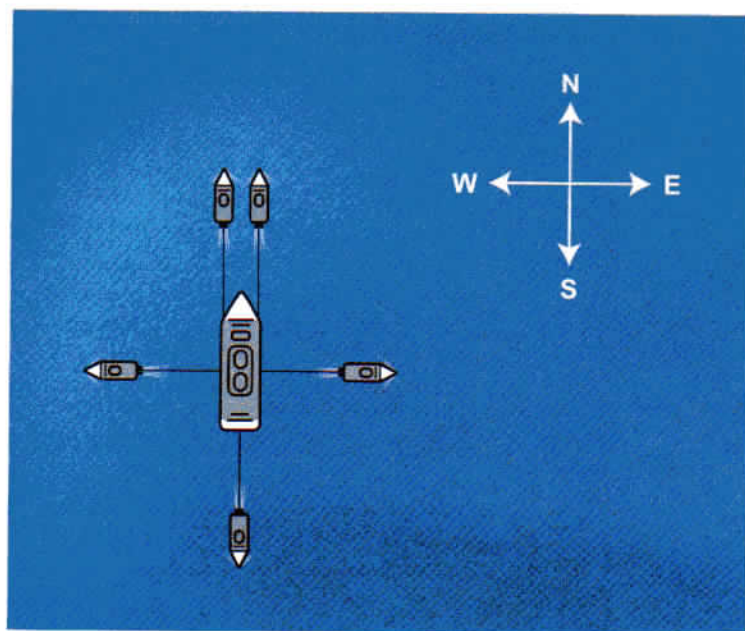
- (A) tin, iron, plastic
 (B) plastic, iron, glass
 (C) iron, aluminium, tin
 (D) glass, wood, plastic
5. The table lists some information about tigers. They are endangered animals, with not many left in the wild or in zoos.

	Type of tiger			
	Siberian	Bengal	Indochinese	Sumatran
Body mass (kg)	90 – 300	136 – 210	115 – 180	90 – 120
Where it lives	Russia	India	Thailand	Sumatra
Number in the wild	360 – 400	3200 – 4600	1200 – 1800	400 – 500
Number in zoos	490	330	60	210

Which two types of tiger have fewer than 2000 living in the wild but more than 100 living in zoos?

- (A) Indochinese and Sumatran
 (B) Siberian and Indochinese
 (C) Siberian and Bengal
 (D) Siberian and Sumatran

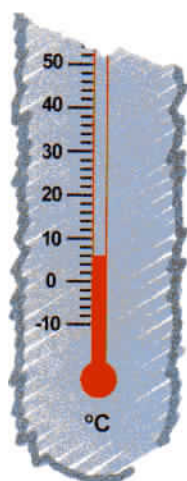
6. The diagram shows a number of tugboats pulling a ship. All forces applied by the tugboats are equally strong.



In which direction will the ship move?

- (A) North
- (B) East
- (C) South
- (D) West

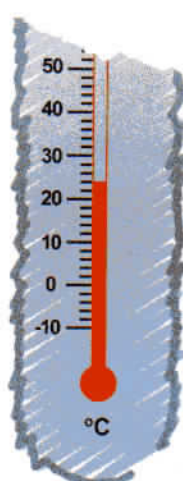
7. Jack used his thermometer to measure the temperature in the school playground at four different times during the day. He recorded his measurements as drawings of his thermometer.



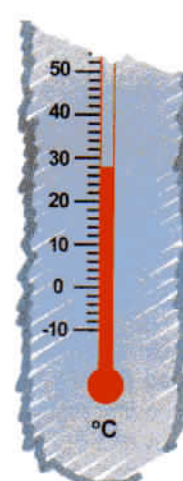
10 am



noon



2 pm



4 pm

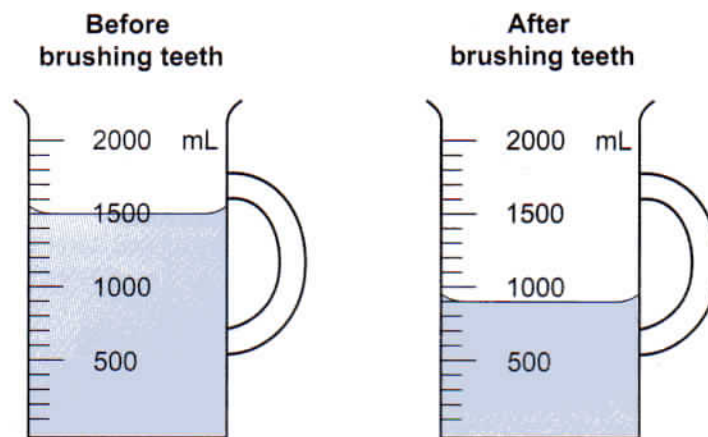
At which two times were the temperature readings the closest to each other?

- (A) 4 pm and 10 am
- (B) noon and 2 pm
- (C) 2 pm and 4 pm
- (D) 10 am and noon

For questions 8 and 9 use the information below.

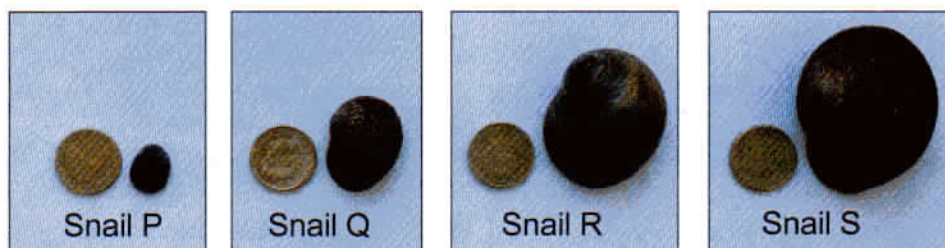
Daniel often leaves the tap running while he brushes his teeth. His parents asked him to find out how much water he actually needs to brush his teeth.

Daniel brushed his teeth by using sips of water from a jug.



8. How many millilitres (mL) of water did Daniel use to brush his teeth?
- (A) 1500 (B) 900
(C) 600 (D) 400
9. Daniel wants to find out how much water he would use if he left the tap running while brushing his teeth. First he used a stopwatch and the jug and measured the amount of water coming out from the tap in 10 seconds.
- What does he need to do next?
- (A) use the stopwatch to measure how long it takes him to brush his teeth
(B) use a kitchen scale to measure the mass of the water he used
(C) use the stopwatch to measure how long it takes to fill the jug
(D) use the jug to measure the water he used in 10 seconds

10. Angela took these photos of four different snail shells, each one next to the same coin.



Max found a snail (T) in the yard and took a photo of it, also next to the same coin.

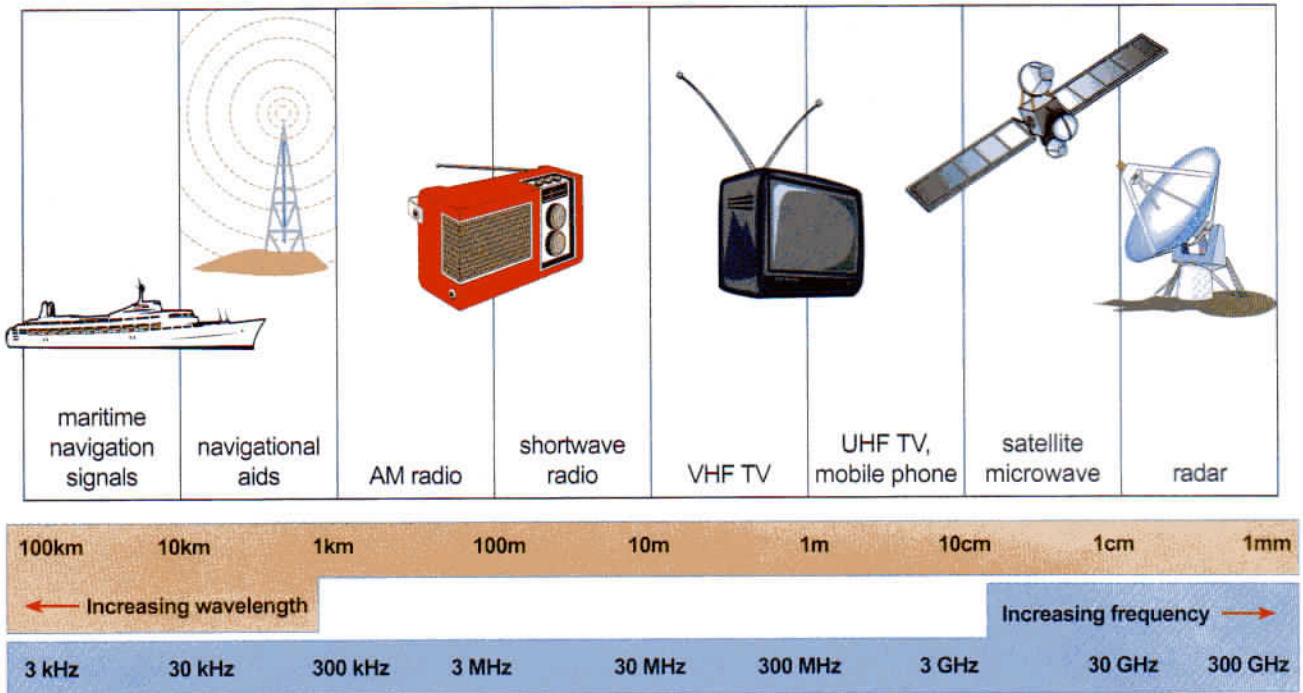


Which snail in Angela's photos is closest in size to Snail T?

- (A) Snail P (B) Snail Q
(C) Snail R (D) Snail S

11. Most forms of electronic communication use electromagnetic waves with a frequency in the radio frequency range.

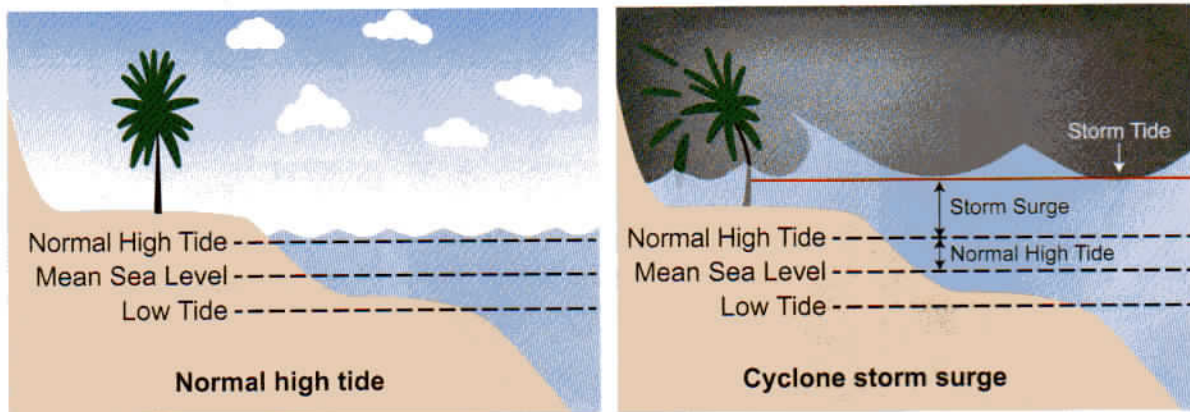
The diagram gives examples of the purpose for which the different frequencies of radio waves are used.



Which electronic device uses radio waves with a wavelength between 1 and 10 metres?

- (A) navigational aids
(B) VHF TV
(C) a mobile phone
(D) a microwave oven

12. The diagrams show the normal high tide water level and the water level during a cyclone storm.



On this island, the normal high tide level is 1 metre above the mean sea level.

Approximately how many metres above the mean sea level is the cyclone storm surge level?

- (A) 0.5
(B) 1.5
(C) 2.5
(D) 3.5

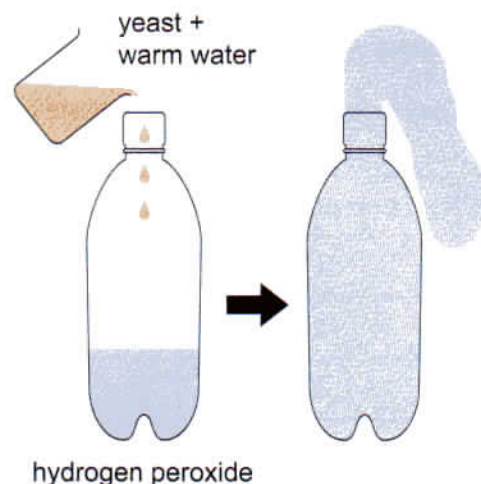
For questions 13 and 14 use the information below.

Max tried to make 'elephant toothpaste' foam based on the instructions in his science book.

He mixed the following chemicals in a bottle:

- 100 millilitres (mL) 6% hydrogen peroxide (liquid)
- 1 packet dry yeast + 80 mL warm water
- 5 mL detergent.

When the chemicals mixed, they started a reaction that produced new chemicals.



13. Which of these must have been produced, that pushed the 'toothpaste' out of the bottle?

(A) gas (B) strawberry milk (C) water (D) detergent

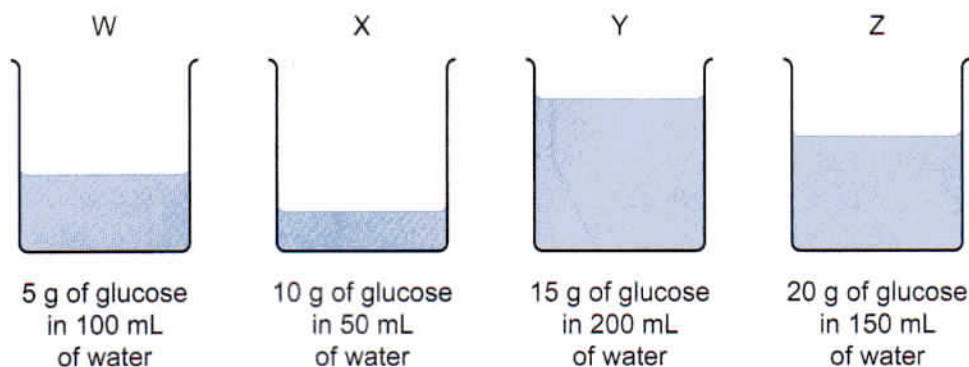
14. Max thought the foam came out too slowly. He repeated the experiment by using 150 mL 6% hydrogen peroxide, 2 packets of dry yeast with 80 mL of warm water and 10 mL of detergent. This time the foam came out a lot faster.

What can Max conclude from this?

- (A) The extra hydrogen peroxide made it faster. (B) The extra yeast made it faster.
(C) The extra detergent made it faster. (D) He cannot conclude what made it faster.

15. Glucose is a type of sugar.

A student dissolved different amounts of glucose in different volumes of water. The water temperature in each container was the same.



Which solution would taste the sweetest?

- (A) W (B) X (C) Y (D) Z

16. Class 4A were learning about disposing of rubbish. The students put their waste either in the compost bin, the recycling bin or the general waste bin.

They recorded the amount of waste in each bin at the end of every week for 4 weeks.

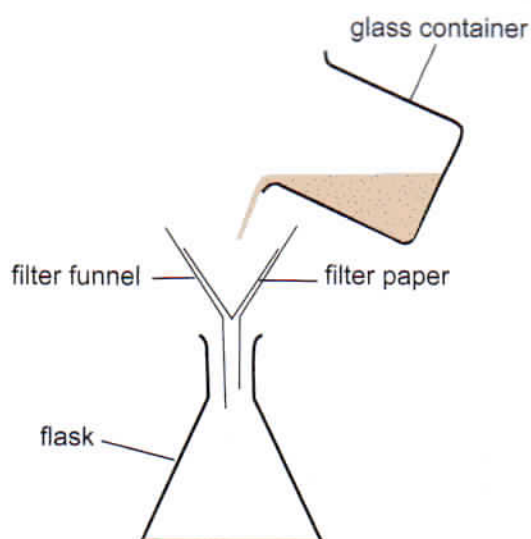
End of Week	Compost (grams)	Recycle bin (items)	General waste (items)
1	40	10	5
2	90	12	6
3	150	15	9
4	205	19	10

During which week did the amount in the compost bin increase the most?

- (A) Week 1
(B) Week 2
(C) Week 3
(D) Week 4

17. Ali poured muddy water onto filter paper inside a filter funnel.

Filter paper only allows water and dissolved substances to pass through it.

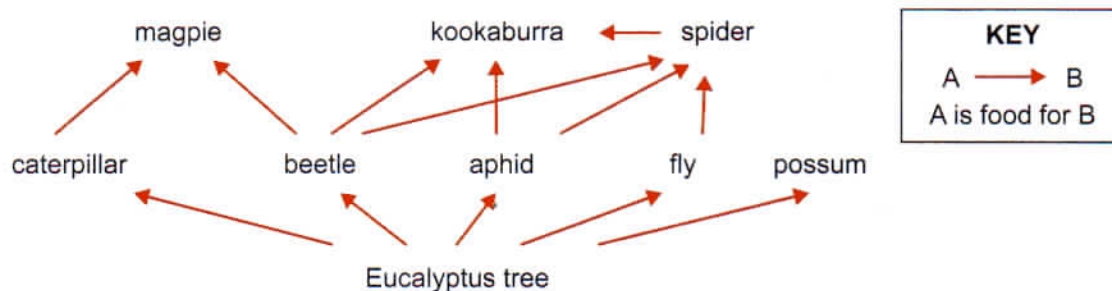


Which option correctly describes the contents of the filter paper and the contents of the flask after Ali finished filtering?

	Contents of filter paper	Contents of flask
(A)	mud	clear water
(B)	muddy water	muddy water
(C)	empty	clear water
(D)	clear water	mud

18. Plants are called producers because they make food using energy from the Sun. An organism that eats another organism is called a consumer. A consumer that eats a producer is called a first-order consumer.

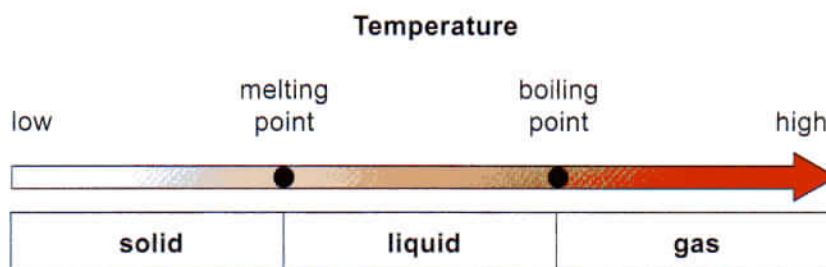
This diagram shows a food web.



How many first-order consumers are shown in this food web?

- (A) 1 (B) 3 (C) 4 (D) 5

19. Things around us are made up of matter. Matter can exist in three main states: solid, liquid or gas. Matter may change state as its temperature changes. The temperature at which a solid becomes a liquid is called its melting point. The temperature at which a liquid becomes a gas is called its boiling point.



Substance X has a melting point of 30 °C and a boiling point of 120 °C.

In what state is substance X at 25 °C and in what state is it at 110 °C?

	at 25 °C	at 110 °C
(A)	solid	liquid
(B)	liquid	gas
(C)	liquid	solid
(D)	solid	gas

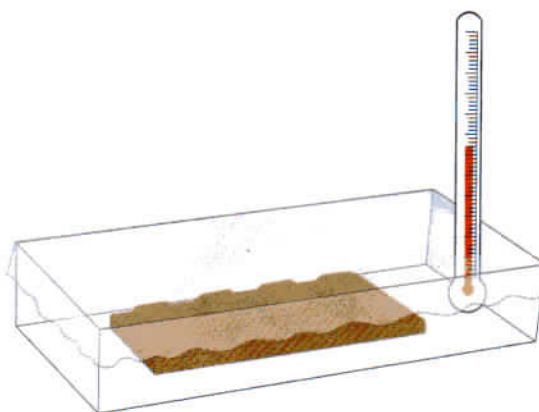
20. Andrew placed a chocolate bar in a clear box outside in the sunlight. He placed a thermometer in the box to measure the temperature inside the box.

He wrote down four statements before and during his investigation.

1. The chocolate will melt at 35 °C.
2. The chocolate lost its crystal structure.
3. The chocolate began to melt at 31 °C.
4. The box trapped heat inside it.

Which statement was an observation?

- | | | | |
|-----|---|-----|---|
| (A) | 1 | (B) | 2 |
| (C) | 3 | (D) | 4 |



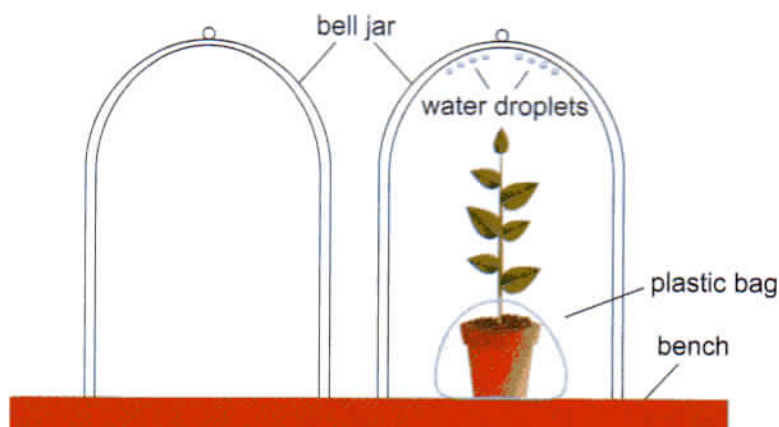
For questions 21 and 22 use the information below.

Liam's teacher told him that plants give off water vapour.

He set up two sealed bell jars. One had a plant in a pot with damp soil in it, the soil covered with a plastic bag.

After 24 hours, he found that the bell jar containing the plant had water droplets on the inside.

Liam compared the bell jar with the plant in it, to an empty bell jar.



21. Why did Liam have to cover the soil in the pot?
- (A) to stop the plant from drying out overnight
 - (B) so that the bell jar would not smell like soil
 - (C) to stop water from the soil interfering with the results
 - (D) so that the droplets on the bell jar could not fall onto the soil
22. What could Liam have put in the empty bell jar to improve the comparison?
- (A) a pot with equally damp, uncovered soil
 - (B) a pot with equally damp soil, covered with a plastic bag
 - (C) a pot with a plant of a different type, set up the same way
 - (D) a pot with the plant covered with plastic and the soil uncovered

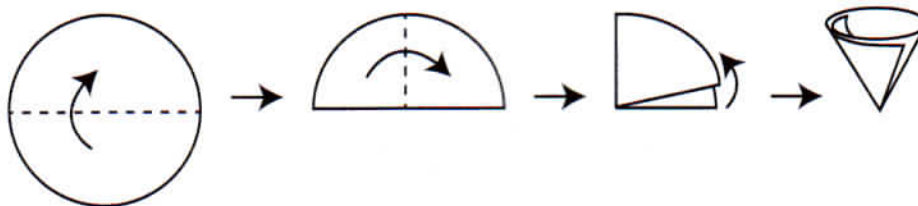
23. The photo shows a small collection of shells. The key helps determine the type of a shell.



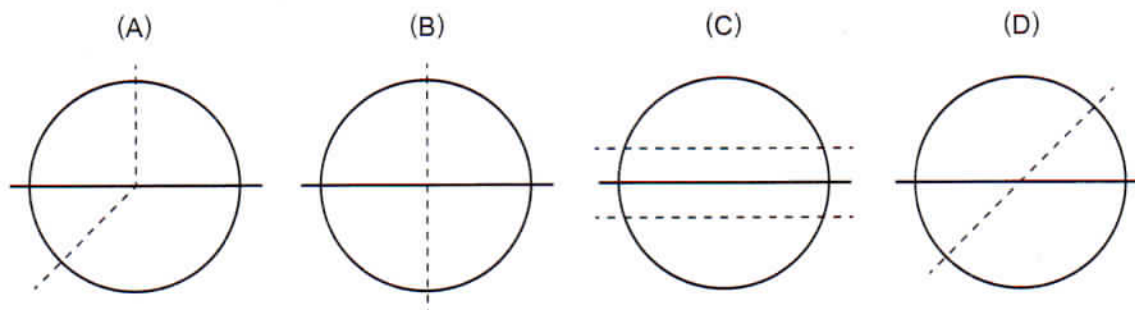
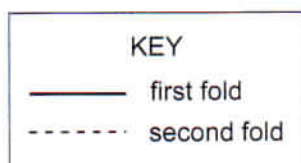
Key to Shells		
i)	a: shell shaped like a snail b: shell not shaped like a snail	Go to iii) Go to ii)
ii)	a: white shell b: black shell	Type P Type Q
iii)	a: shell with stripes or lines b: shell without stripes or lines	Go to iv) Go to v)
iv)	a: black stripes or lines b: brown stripes or lines	Type R Type S
v)	a: shell with square markings b: shell without square markings	Type T Type U

According to the table, what type of shells are the two shells circled in the photo?

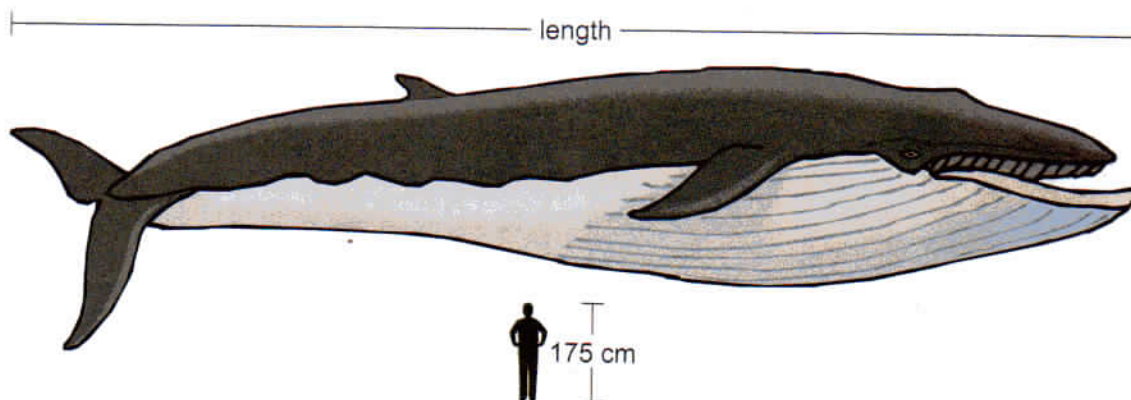
- (A) Type S (B) Type T (C) Type R (D) Type U
24. Filter paper is a special paper used to separate solids from liquids. The drawings show how to fold a piece of filter paper.



Which diagram shows the lines along which the paper has been folded?



25. The Blue Whale, a marine mammal, is the largest animal known to have lived on Earth.



The man and the Blue Whale are drawn to the same scale.

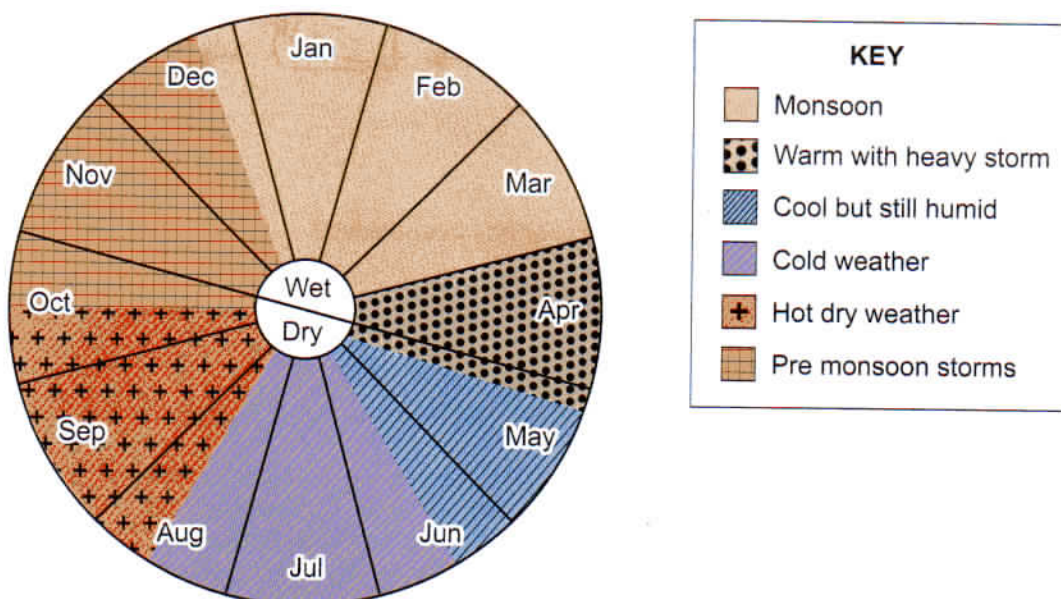
What is the approximate length of the Blue Whale?

- (A) 20 m (B) 30 m (C) 40 m (D) 50 m

For questions 26 and 27 use the information below.

Aboriginal people in Australia used fire to help them manage the land. They burnt the woodlands when the weather was cool and dry. Carefully managed small and regular burns significantly reduced the intensity of the fires started by lightning during the dry weather.

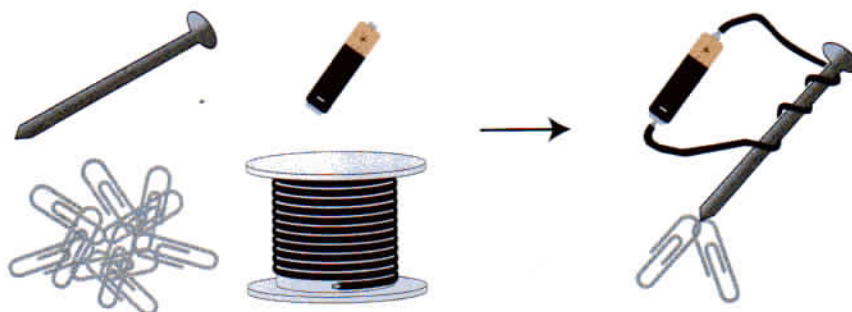
The calendar below shows the six different seasons of a year in the Kakadu region in northern Australia.



26. In what months did Aboriginal people in Kakadu use fire to manage the land?
- (A) February – May (B) June – August
(C) August – November (D) November – February
27. Why do the small and regular burns reduce the intensity of fires in the dry weather?
- (A) there are not as many plants to burn in the dry weather.
(B) there is not much fire left to burn after the small burns.
(C) the cold weather puts out the fire.
(D) small burns remove all the grass.

28. Xiao Ming read in a book that electricity can be used to turn a piece of iron into a magnet. He decided to test this idea.

He carried out his investigation using a large iron nail, some plastic-coated wire, a brand new battery and some paperclips. These drawings show what he did.



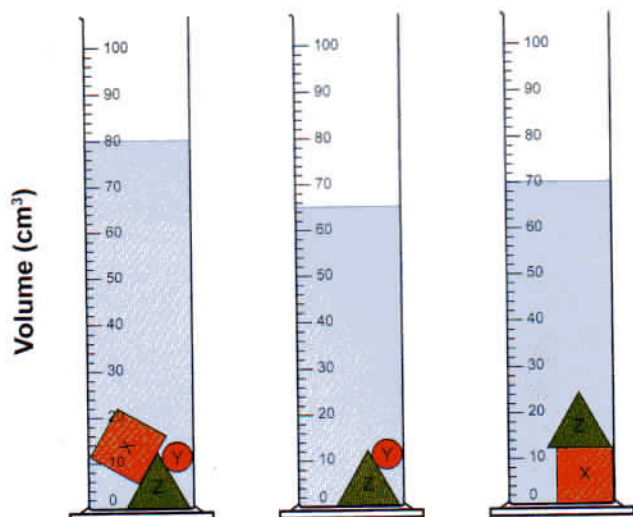
Xiao Ming wanted to carry out another investigation with the materials he had.

Which of these questions would he be able to answer with this investigation?

- (A) Does the size of the iron nail affect the strength of the magnet?
- (B) Does the number of loops of wire around the nail affect the strength of the magnet?
- (C) Does the thickness of the wire affect the strength of the magnet?
- (D) Does the power of the battery affect the strength of the magnet?

29. The volume of an object is equal to the volume of water it displaces.

Jason set up an investigation to find the volume of several objects using measuring cylinders and 50 mL of water in each measuring cylinder.



What are the volumes of objects X, Y and Z?

	Volume (cm ³)		
	X	Y	Z
(A)	5	10	55
(B)	10	15	5
(C)	15	5	5
(D)	15	10	5

For question 30 and 31 use the information below.

A spring balance can be used to measure the mass and the weight of an object. It measures mass in grams and weight in newtons.

Ali used a spring balance to find the mass of some keys.



The table shows his results.

Key(s)	Mass (grams)
1	14
1+2	27
1+2+3	42
1+2+3+4	57

30. Which two keys have the same mass?

- (A) Key 1 and Key 2
(C) Key 2 and Key 4

- (B) Key 2 and Key 3
(D) Key 3 and Key 4

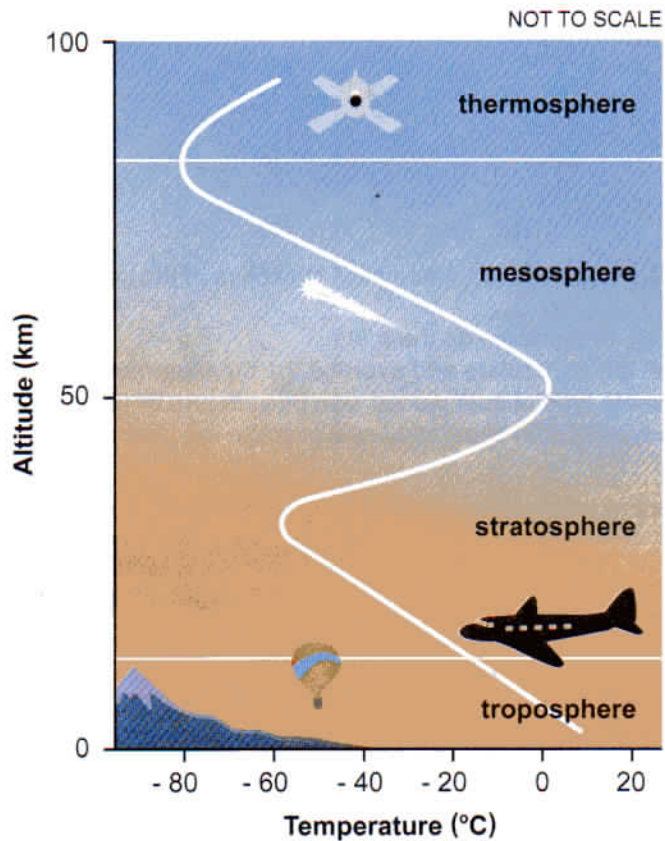
31. Ali found the weight of his pencil case to be 5 newtons.

What is the mass of his pencil case, in grams?

- (A) 500
(C) 5

- (B) 50
(D) 0.5

32. The blanket of gases that surrounds Earth is called the atmosphere. It is divided into four layers. The curved line in the diagram represents the temperature at different altitudes (height above sea level) in the atmosphere.



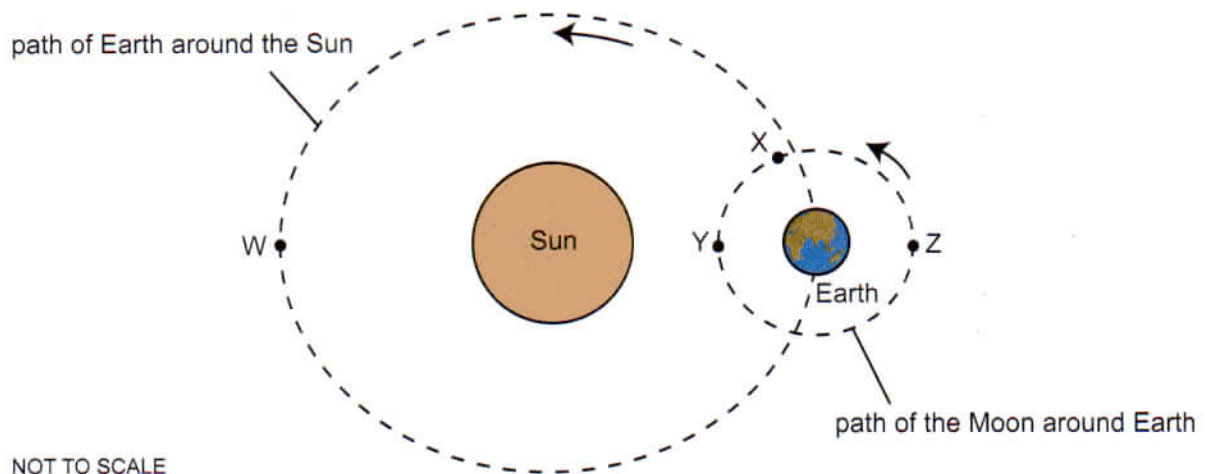
Which statement is correct, based on the diagram?

- (A) Within the troposphere, the temperature increases with altitude.
- (B) As altitude increases, the temperature steadily increases.
- (C) Within the stratosphere, temperature increases as the altitude increases.
- (D) The highest temperature in the mesosphere is about 0°C .

33. Nearly all the light that reaches Earth comes from the Sun.

A lunar eclipse occurs when the Moon moves into Earth's shadow.

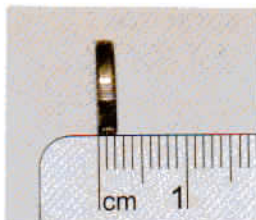
The diagram represents the path of Earth and the Moon as they move around the Sun together.



In which location shown on the diagram is the Moon during a lunar eclipse?

- (A) W (B) X (C) Y (D) Z

34. Jason used a ruler to measure the thickness of a coin. He recorded his measurement as "about two millimetres". Jason's teacher asked him to make a more accurate estimate.

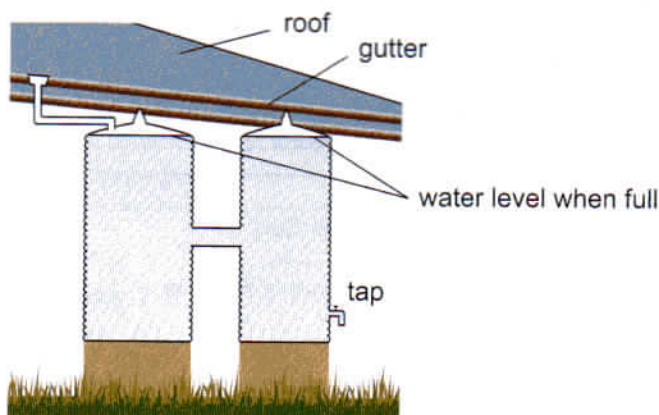


Which of the following is the best way to get a more accurate estimate of the thickness of the coin?

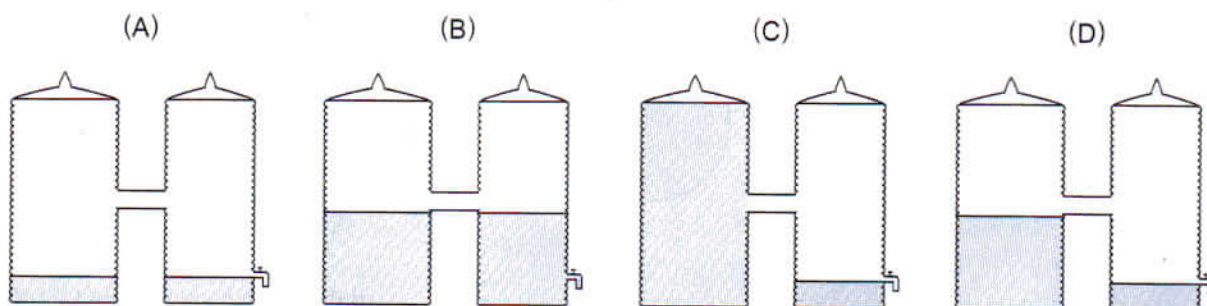
- (A) use a magnifying glass to look more closely at the scale on the ruler
- (B) measure the thickness of the coin with ten different rulers and average the measurements
- (C) measure the total thickness of 10 of the same type of coin placed side by side and divide by 10
- (D) measure the thickness of 10 of the same type of coin separately and average the measurements

35. Sylvio had two rainwater tanks connected by a pipe as shown. The second tank has a tap close to the bottom so he can use the water on the garden. Both tanks were full.

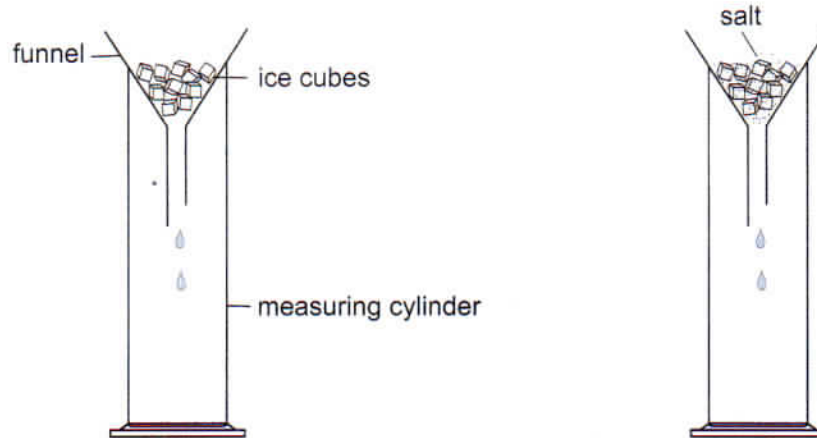
One day, the tap was accidentally left open and the water poured out until it stopped.



What was the water level in the two tanks when the water stopped pouring out?

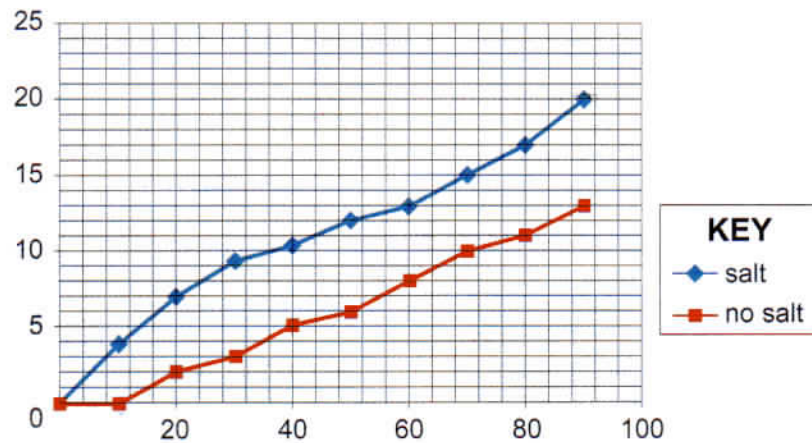


36. Lana investigated the effect of salt on melting ice. She placed equal amounts of ice in two funnels. She added salt to the ice in one funnel.



She placed the measuring cylinders close together in the classroom and measured the amount of water in each cylinder at 10-minute intervals.

The graph represents some of her results.

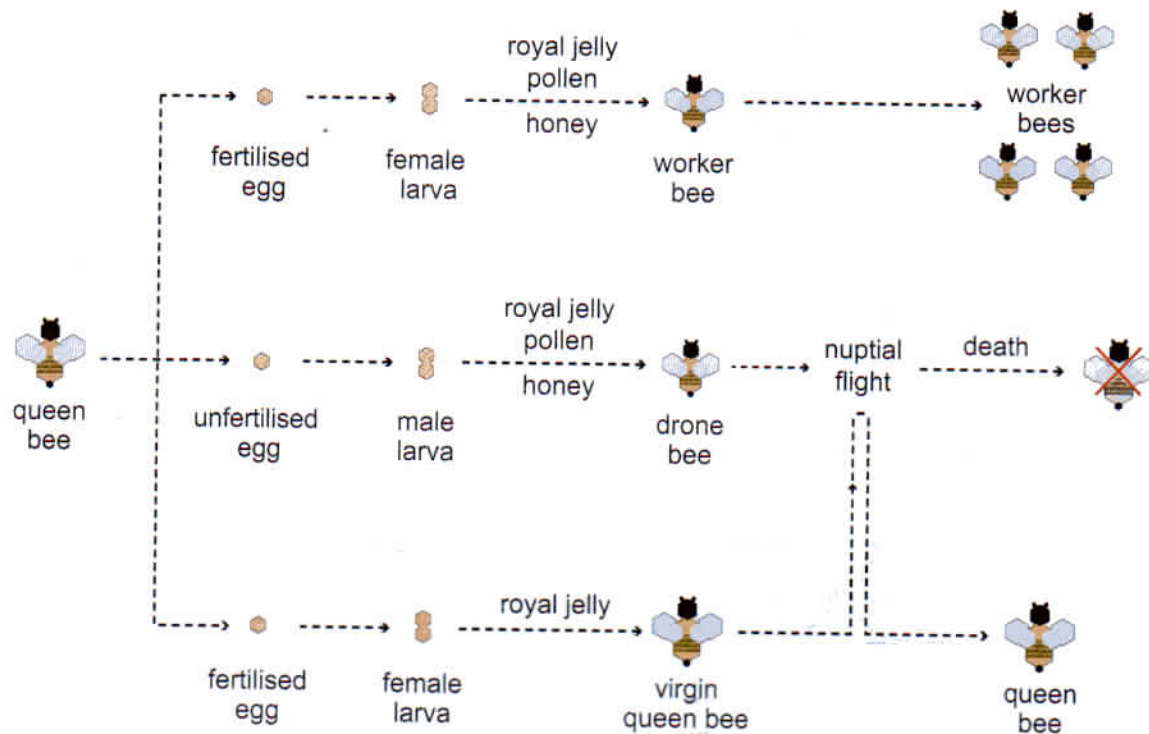


What can Lana conclude from her results?

- (A) Ice needs salt to melt. (B) Salt makes ice melt faster.
(C) Ice melts faster without salt. (D) The more salt the faster the ice will melt.

For question 37 and 38 use the information below.

The diagram shows the life cycle of honeybees.



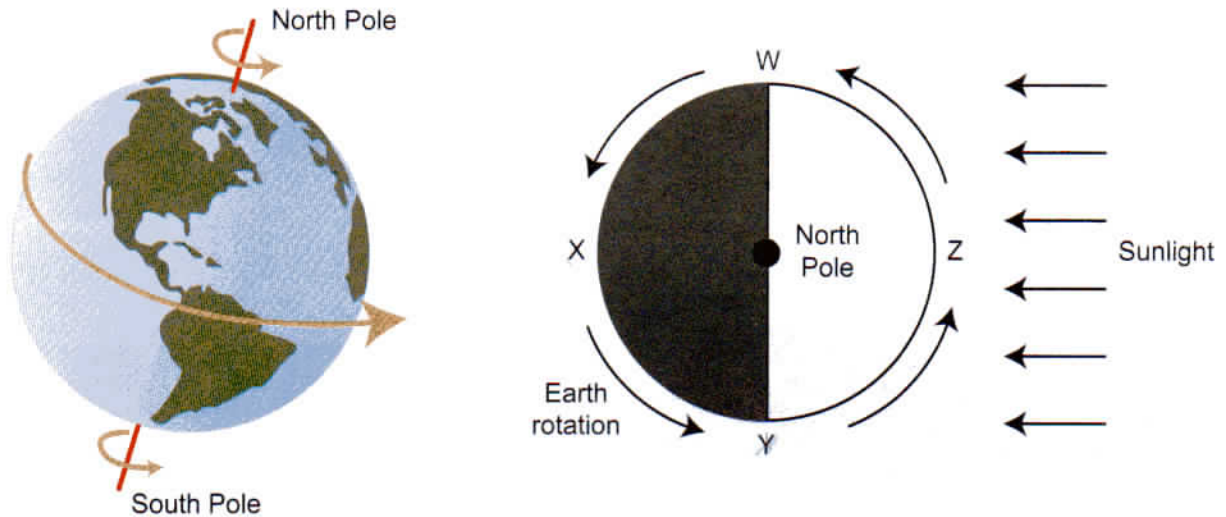
37. Which of the statements is correct according to the diagram?

- (A) A queen bee produces female bees only.
- (B) A queen bee can turn into a worker bee.
- (C) A virgin queen bee mates with a drone bee to become a queen bee.
- (D) A drone bee would become a queen bee if it did not die after the nuptial flight.

38. Which of the following determines whether an egg will become a drone bee or a worker bee?

- (A) whether the larva is male or female
- (B) whether the egg is fertilised or not
- (C) whether it goes on a nuptial flight
- (D) whether the larva is fed pollen and honey

39. Earth rotates about its axis from west to east. On the side of Earth facing the sun it is day time while on the other side it is night time.

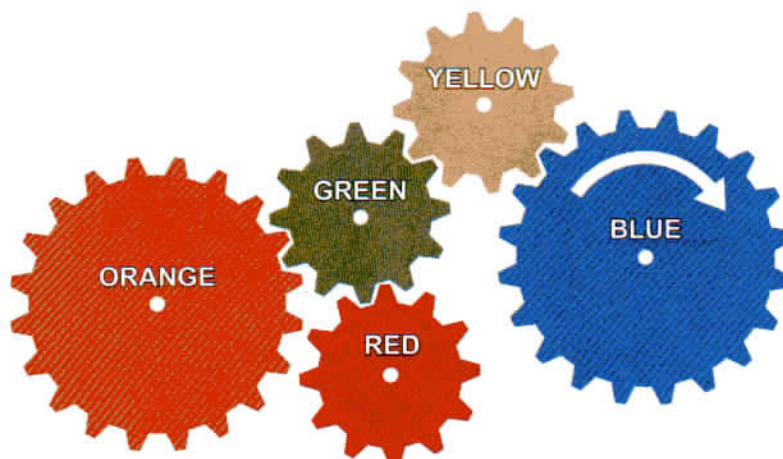


Which of the following correctly identifies the time of the day at W, X, Y and Z?

	Sunset	Sunrise	Noon	Midnight
(A)	Y	X	W	Z
(B)	W	Y	Z	X
(C)	W	Y	X	Z
(D)	Y	W	Z	X

40. A cog is a wheel with teeth around its edge. When cogs 'mesh', their teeth fit together and one cog can make another cog turn without slipping.

Alice set up a set of cogs as shown in the diagram.



She turned the blue cog in the direction shown by the arrow.

Which way did the red cog turn?

- | | |
|---|--|
| (A) in the same direction, more quickly | (B) in the same direction, more slowly |
| (C) in the opposite direction, more quickly | (D) in the opposite direction, more slowly |

Acknowledgment

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Sources

Questions 37–38 Life cycle diagram adapted from a diagram by Cecilian91 <https://commons.wikimedia.org/wiki/File:Honey_Bees_-_Sexes_and_castes.svg> licensed under the Creative Commons Attribution-ShareAlike 4.0 International licence <<https://creativecommons.org/licenses/by-sa/4.0/>>.

The following year levels should sit THIS Paper:

Australia	Year 5
Brunei	Primary 5
Indonesia	Year 6
Malaysia	Standard 5
New Zealand	Year 6
Pacific Region	Year 6
Singapore	Primary 4
South Africa	Grade 5



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2016 SCIENCE ANSWER KEYS

QUESTION NUMBER	PAPER A	PAPER B	PAPER C	PAPER D	PAPER E	PAPER F	PAPER G	PAPER H	PAPERS I & J
1	A	D	A	A	A	C	A	D	B
2	C	A	C	C	B	A	C	B	A
3	A	D	B	B	D	B	C	A	C
4	B	B	D	C	B	C	D	A	D
5	B	A	D	D	C	C	A	B	B
6	C	D	A	A	C	B	D	A	B
7	A	A	C	C	A	C	B	D	C
8	C	B	C	D	C	D	B	D	B
9	D	B	A	C	D	C	D	C	A
10	A	C	C	C	A	B	A	D	C
11	C	C	B	A	C	C	C	C	D
12	B	D	C	D	B	B	B	A	A
13	C	C	A	D	C	D	B	C	C
14	D	C	D	B	C	B	D	A	B
15	D	D	B	A	C	A	A	B	C
16	A	B	C	C	B	B	A	B	C
17	D	C	A	C	A	A	B	D	D
18	B	A	D	B	A	A	D	B	B
19	D	A	A	B	D	D	D	C	A
20	A	C	C	A	C	B	A	C	D
21	B	B	C	D	B	A	B	D	C
22	C	A	B	B	B	C	C	A	D
23	B	B	D	B	C	D	B	D	C
24	A	B	B	A	C	D	C	B	C
25	B	D	A	D	C	B	D	B	A
26	A	C	B	B	D	D	B	C	B
27	D	C	A	C	C	B	A	C	A
28	C	B	B	A	A	D	C	C	A
29	D	A	D	A	C	C	D	D	C
30	C	D	D	D	D	A	B	A	B
31			A	D	B	C	A	A	B
32			D	C	A	D	C	D	A
33			D	D	B	B	B	B	C

2016 SCIENCE ANSWER KEYS

QUESTION NUMBER	PAPER A	PAPER B	PAPER C	PAPER D	PAPER E	PAPER F	PAPER G	PAPER H	PAPERS I & J
34			C	D	A	A	D	B	B
35			D	D	D	D	C	B	A
36			B	C	C	D	C	C	D
37			C	B	B	C	A	D	A
38			B	A	D	B	A	D	D
39			B	B	D	A	C	C	D
40			C	B	D	A	D	A	B
41					C	C	D	B	D
42					A	C	B	B	A
43					D	D	D	B	C
44					A	A	B	A	B
45					A	B	C	D	A

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